Lecture 9.

Suggested Reading
3.1-3.2.

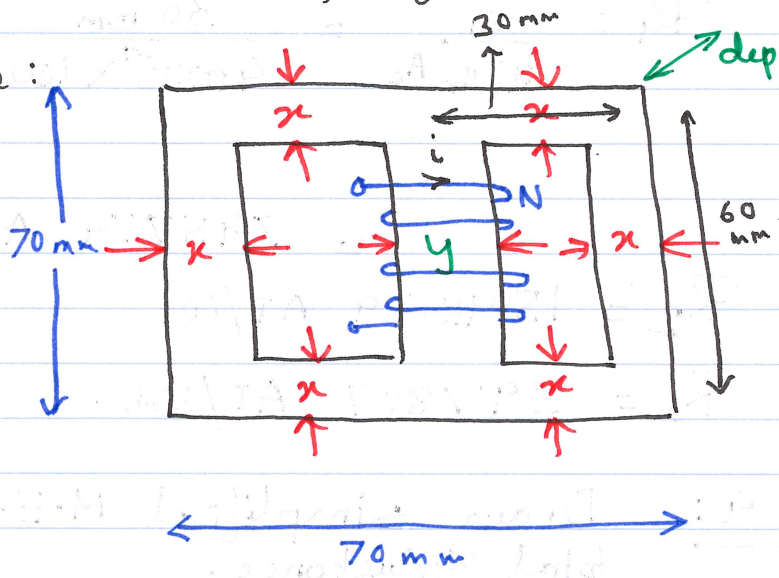
Last time : * Magnetic circuits.
* Proved : $v = L\frac{di}{dt}$ using Maxwell's equation.

$$\downarrow$$

$$\mathcal{R} = \{ f(\text{geometry, material property}) \quad \frac{N^2}{\mathcal{R}}$$

Today : * Use magnetic equivalent circuits
* to solve for flux density, inductance.

Example :



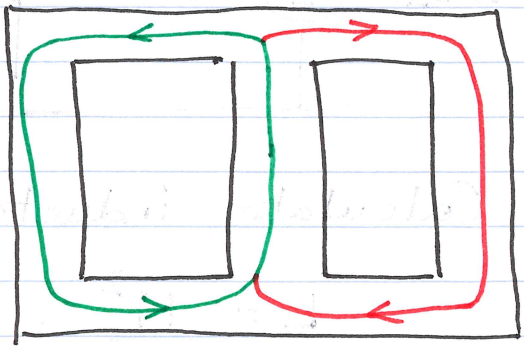
$x = 10$ mm
$y = 20$ mm

$\mu_r = 100$ (ferrite).

$N = 50$ turns.

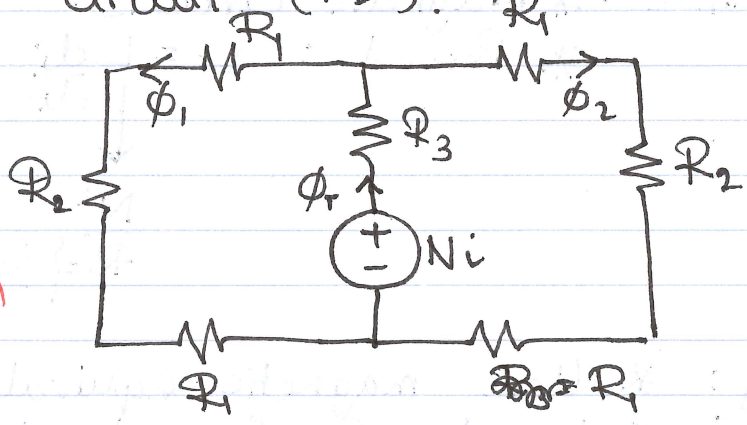
Calculate inductance and flux densities.

Step 1 : Identify flux path. $B = \mu_0 \mu_r H$.



We will assume as if the flux is flowing through the center. (mean length path).

Step 2: Draw the magnetic equivalent circuit. (MEC).



$\phi_T = \phi_1 + \phi_2$

$R = \frac{l}{\mu_0 \mu_r A}$ ✓ ✓ ✓

If any of (l, μ_r, A) these changes, we need to introduce a reluctance.

$$\mathcal{R}_1 = \frac{l_1}{\mu_0 \mu_r A_c} = \frac{30\text{ mm}}{4\pi \times 10^{-7} \times 100 \times 10\text{ mm} \times 30\text{ mm}}$$

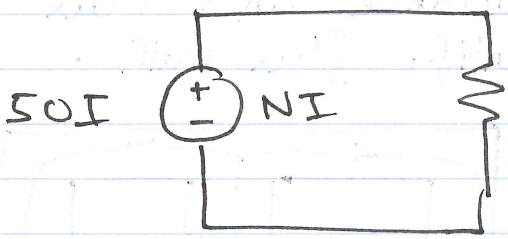
(30 mm: ↑ depth)

$$= 795774 \text{ AT/Wb}.$$

$$\mathcal{R}_2 = 159549 \text{ AT/Wb}.$$

$$\mathcal{R}_3 = 397887 \text{ AT/Wb}.$$

Step 4: Draw simplified M.E.C, calculate total Reluctance.



$$\mathcal{R}_T = \mathcal{R}_3 + \frac{2\mathcal{R}_1 + \mathcal{R}_2}{2}$$

Step 5: Calculate inductance,

$$L = \frac{N^2}{\mathcal{R}_T} = 1.05 \text{ mH}.$$

From the measurement : 0.98 mH.

% error = 2 %. [pretty accurate]. even without solving full fledged Maxwell's equation.]

How about flux density ?

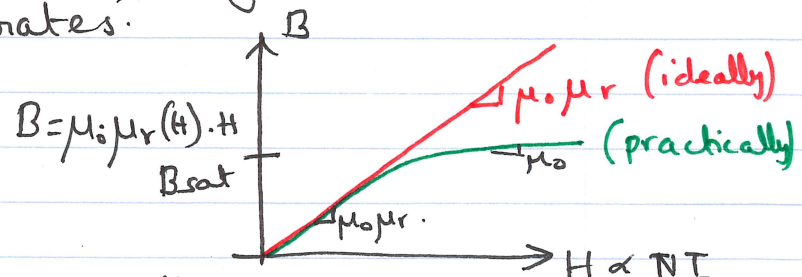
Calculate ϕ_1 and ϕ_2.

$\therefore$ $B_{\text{center-limb}}$ (flux density) $= \dfrac{\phi_T}{A_3}$.

$B_{\text{side-limb}} = \dfrac{\phi_1}{A_1}$.

Why important ?

Ideally does not matter.
Practically, magnetic material saturates.



For ferrite, $B_{sat} = 0.3$ T.

So, if $B_{calc} > B_{sat}$

$\Downarrow$

$\mu_r \neq 100$

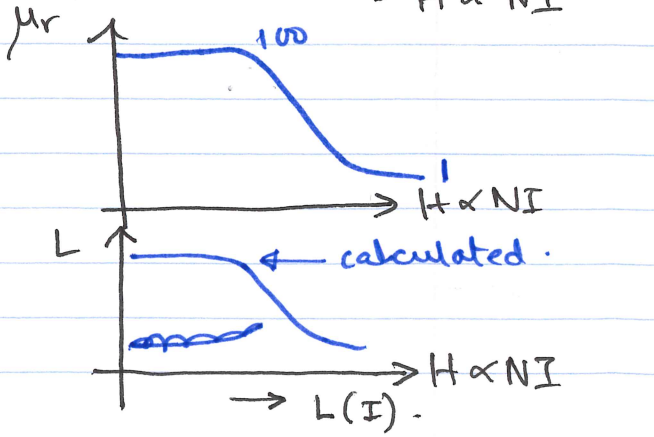
$\Downarrow$

R changes

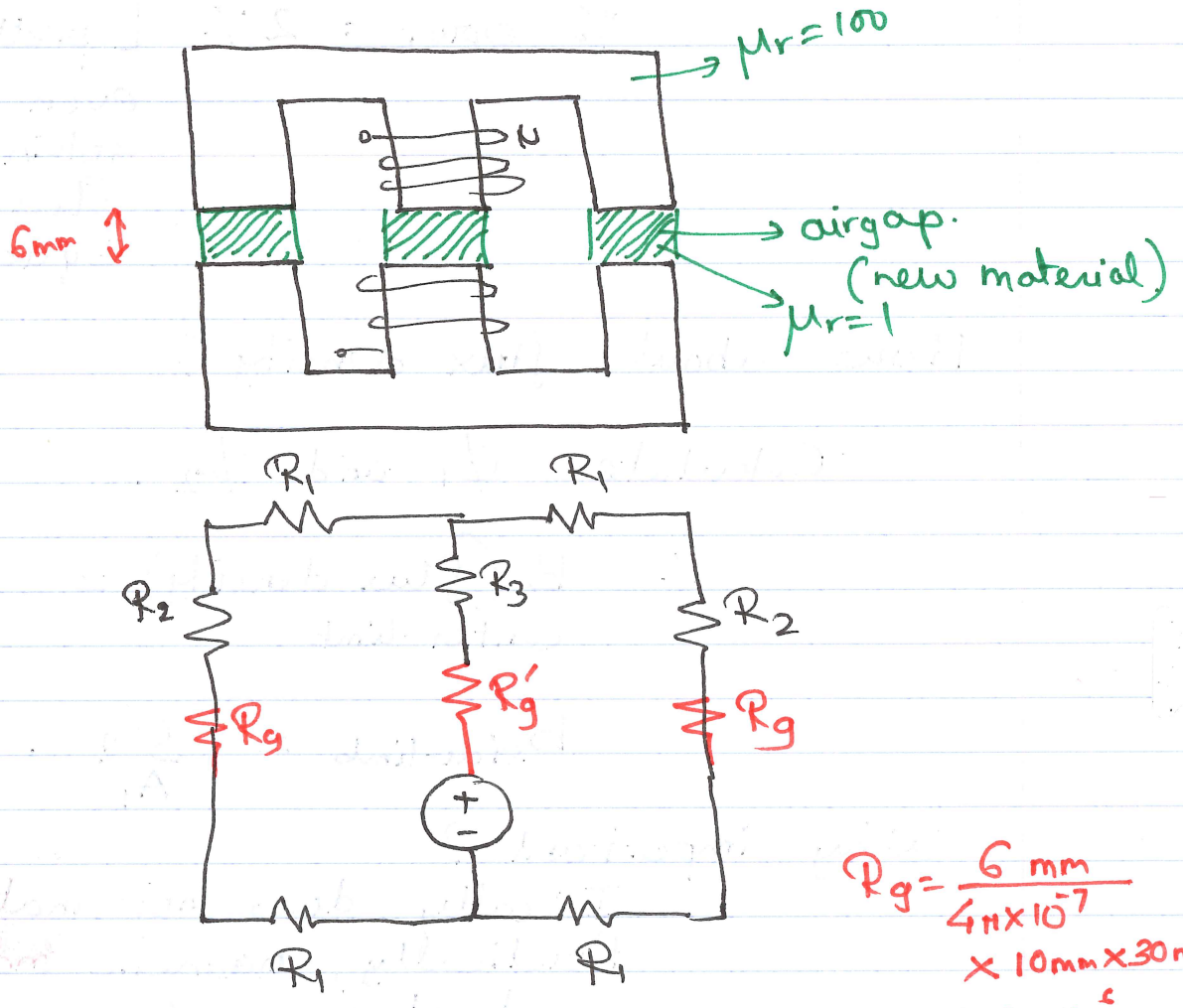
$\Downarrow$

L changes

$R = \frac{l}{\mu A}$



Let us introduce ~~a~~ airgap. [6 mm block.]



$$R_g = \frac{6\text{ mm}}{4\pi \times 10^{-7} \times 10\text{mm} \times 30\text{mm}}$$

$$= 15.91 \times 10^6 \text{ AT/wb}$$

$$R_g' = \frac{6\text{ mm}}{4\pi \times 10^{-7} \times 20\text{mm} \times 30\text{m}}$$

$$= 7.96 \times 10^6 \text{ AT/wb.}$$

$$\therefore \quad R_T\big|_{New} = 18.30 \times 10^6 \text{ AT/wb.}$$

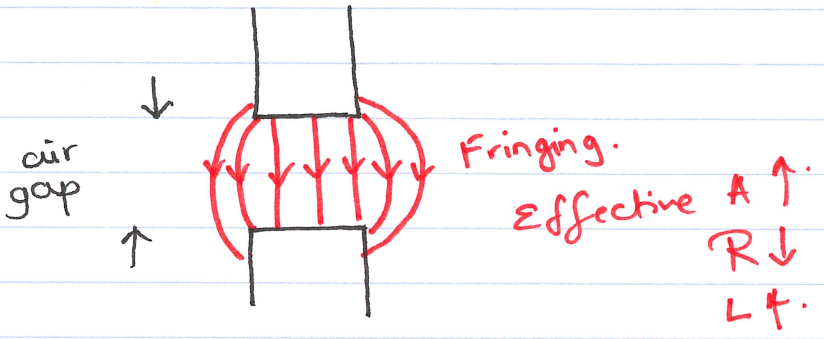
VS

$$R_T\big|_{old} = 2.38 \times 10^6 \text{ AT/wb.}$$

$L_{est} = 139\ \mu H$ (low as expected).

$L_{measured} = 313\ \mu H$

% error: 56 % ⟵ ??



Something to think:

What is the advantage of adding a gap?